

Introduction to Linear Algebra

Vectors, Matrices, Systems, and Geometry

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AI and AIS Summer School July/Aug 2026

What Is Linear Algebra?

Many systems involve several quantities at once:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Linear algebra provides the mathematical tools for working with such collections of quantities.

It studies:

- vectors containing several components,
- equations involving several unknowns,
- matrices that transform vectors,
- geometric ideas such as direction and angle.

This structure allows complex, high-dimensional problems to be represented and solved systematically.

Vector Spaces: Vectors Live in a Coordinate Space

The coordinate space $\mathbb{R}^n$ is the set of all ordered lists of n real numbers:

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \right\}.$$

A vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

is one element of this space.

Dimension of the ambient space

$\mathbb{R}^2$: vectors with two components

$\mathbb{R}^3$: vectors with three components

$\mathbb{R}^n$: vectors with n components

The components may represent:

- spatial coordinates,
- measured signals,
- the state of a dynamical system.

Vectors

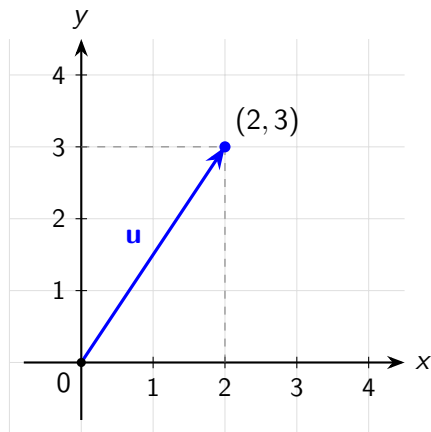
A vector is an ordered collection of numbers:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Geometrically, it represents a displacement:

- 2 units in the x -direction,
- 3 units in the y -direction.

The vector is drawn from the origin to the point $(2, 3)$.



Scalar Multiplication

Multiplying a vector by a scalar changes its length:

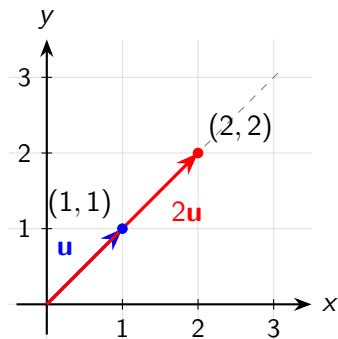
$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For example,

$$2\mathbf{u} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The vector $2\mathbf{u}$:

- points in the same direction,
- has twice the length.



$$\mathbf{u} \longrightarrow 2\mathbf{u}$$

Vector Addition

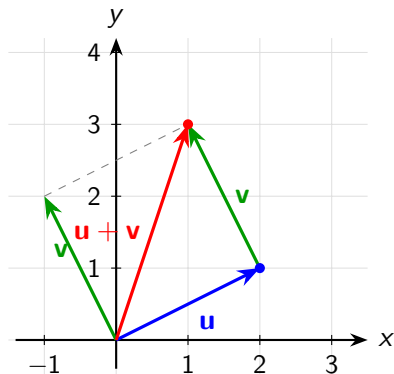
Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 2 + (-1) \\ 1 + 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Geometrically, place the tail of $\mathbf{v}$ at the head of $\mathbf{u}$.



Linear Combinations

A linear combination has the form

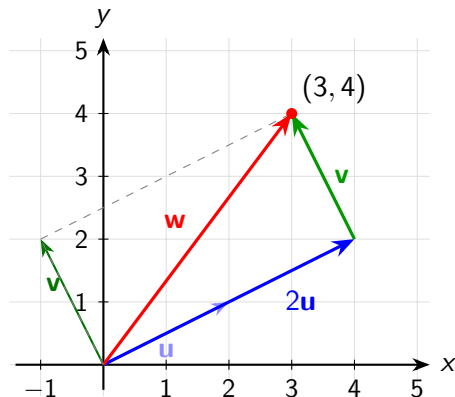
$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$

Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Using $a = 2$ and $b = 1$,

$$\begin{aligned} \mathbf{w} &= 2\mathbf{u} + \mathbf{v} \\ &= 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \end{aligned}$$



$$\mathbf{w} = 2\mathbf{u} + \mathbf{v}$$

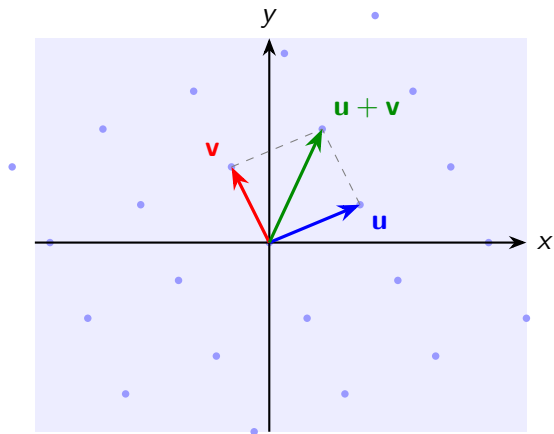
Span: Independent Vectors

The span of two vectors is the set of all their linear combinations:

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \{a\mathbf{u} + b\mathbf{v} : a, b \in \mathbb{R}\}.$$

If $\mathbf{u}$ and $\mathbf{v}$ point in different directions, changing a and b allows us to reach every point in the plane.

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2.$$



Independent directions span the plane

Span: Dependent Vectors

Suppose one vector is a multiple of the other:

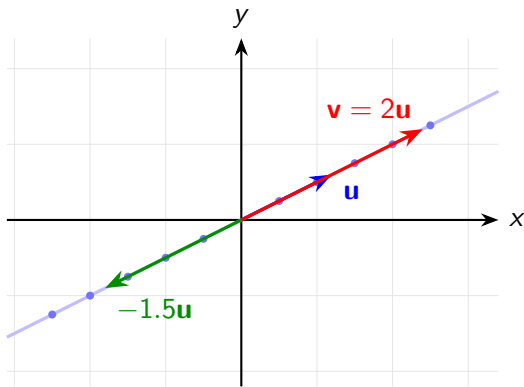
$$\mathbf{v} = 2\mathbf{u}.$$

Then

$$a\mathbf{u} + b\mathbf{v} = a\mathbf{u} + 2b\mathbf{u} = (a + 2b)\mathbf{u}.$$

Every linear combination still points along the same direction.

Therefore, the vectors span only a line rather than the whole plane.



Dependent vectors span only a line

Span: Two Examples

Example 1:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

span all of $\mathbb{R}^2$.

Example 2:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

span only a line because

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Linear Independence

Vectors are linearly independent if none of them can be constructed from the others.

For two vectors, we test:

$$c_1 \mathbf{u} + c_2 \mathbf{v} = \mathbf{0}.$$

If the only solution is

$$c_1 = 0, \quad c_2 = 0,$$

then the vectors are linearly independent.

If there is a nonzero solution, the vectors are linearly dependent.

Basis

A basis is a set of vectors that satisfies two conditions:

- 1 The vectors span the space.
- 2 The vectors are linearly independent.

The standard basis of $\mathbb{R}^2$ is

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Any vector can be written as

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2.$$

Subspaces of $\mathbb{R}^n$

Let

$$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n.$$

Their span is the set of all linear combinations:

$$W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k : a_i \in \mathbb{R}\}.$$

The set W is a **subspace** of $\mathbb{R}^n$.

Subspace property

If $\mathbf{x}, \mathbf{y} \in W$ and $a, b \in \mathbb{R}$, then

$$a\mathbf{x} + b\mathbf{y} \in W.$$

Thus, a subspace is closed under addition and scalar multiplication.

Independent Vectors Determine Dimension

Suppose that

$$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$$

are linearly independent.

Then

$$W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$$

is a k -dimensional subspace:

$$\dim(W) = k.$$

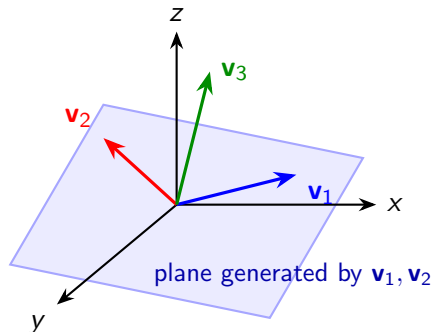
Key result

k linearly independent vectors span a k -dimensional subspace.

Subspaces in $\mathbb{R}^3$

In $\mathbb{R}^3$:

- One nonzero vector spans a line.
- Two linearly independent vectors span a plane.
- Three linearly independent vectors span all of $\mathbb{R}^3$.



Independent vectors	Subspace
1	line
2	plane
3	$\mathbb{R}^3$

$$\mathbf{u} = \mathbf{p} + s\mathbf{v}_1 + t\mathbf{v}_2, \quad \mathbf{p} \in \mathbb{R}^3, \quad s, t \in \mathbb{R}.$$

$\mathbf{p}$ is a fixed point and s and t are variables

When Do Vectors Span the Whole Space?

The space $\mathbb{R}^n$ has dimension n .

Therefore:

- If $k < n$, then k linearly independent vectors span a k -dimensional proper subspace of $\mathbb{R}^n$.
- If $k = n$, then n linearly independent vectors span all of $\mathbb{R}^n$.
- More than n vectors in $\mathbb{R}^n$ cannot all be linearly independent.

Basis

A set of vectors is a basis of a space if it is

linearly independent and spans the space.

Hence, any n linearly independent vectors in $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$.

Equivalent Results in $\mathbb{R}^n$

For a collection of exactly n vectors in $\mathbb{R}^n$, the following statements are equivalent:

- 1 The vectors are linearly independent.
- 2 The vectors span $\mathbb{R}^n$.
- 3 The vectors form a basis of $\mathbb{R}^n$.

Thus, for exactly n vectors, proving either independence or spanning is sufficient.

$$n \text{ independent vectors in } \mathbb{R}^n \iff \text{a basis of } \mathbb{R}^n$$

Linear Equations as Geometry

Solving linear equations has a geometric meaning.

Equation type	Geometric object
One equation in two unknowns	line in $\mathbb{R}^2$
One equation in three unknowns	plane in $\mathbb{R}^3$
Several equations	intersections of lines or planes

The solution set is the set of points satisfying all equations at once.

Solving equations means finding where geometric objects intersect.

One Linear Equation in Two Unknowns: A Line

Consider

$$x + y = 4.$$

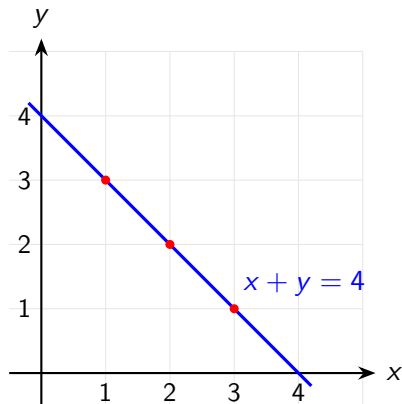
This equation does not determine a unique point.

Instead, it describes all points (x, y) whose coordinates add to 4.

$$(1, 3), \quad (2, 2), \quad (3, 1)$$

all satisfy the equation.

One equation in two unknowns with nonzero coefficients gives a line in $\mathbb{R}^2$



Two Equations in Two Unknowns: Intersecting Lines

Now consider

$$x + y = 4,$$

$$x - y = 1.$$

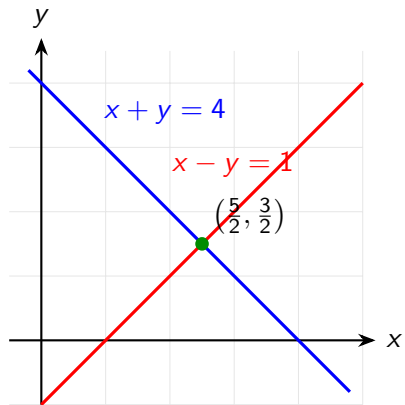
Each equation is a line.

A solution must lie on both lines.

The intersection point is

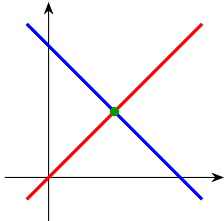
$$x = \frac{5}{2}, \quad y = \frac{3}{2}.$$

So this system has one solution.



What Can Happen in Two Dimensions?

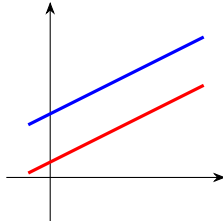
One solution



intersecting lines

Two independent constraints.

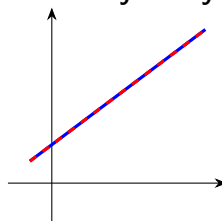
No solution



parallel lines

The constraints conflict.

Infinitely many



same line

One equation repeats the other.

One Equation in Three Unknowns: A Plane

Consider

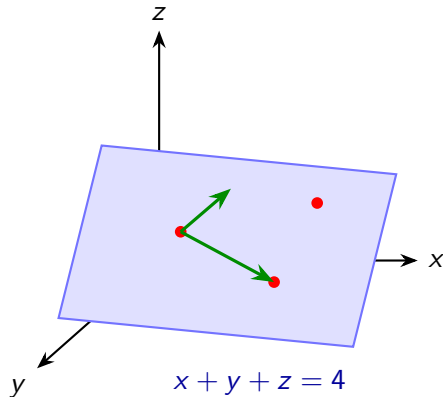
$$x + y + z = 4.$$

This is one equation with three unknowns.

It does not determine a unique point.

Instead, it describes a plane in $\mathbb{R}^3$.

There are two free directions of motion on the plane.

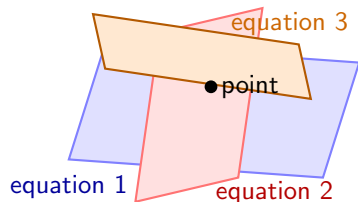


Several Equations in Three Unknowns

In $\mathbb{R}^3$, each independent linear equation usually removes one degree of freedom.

Independent equations	Typical solution set
1	plane
2	line
3	point

More equations can be added, but in three variables there can be at most three independent constraints.



Consistent and Inconsistent Systems I

A system is **consistent** if all equations can be satisfied simultaneously.

It is **inconsistent** if the constraints conflict.

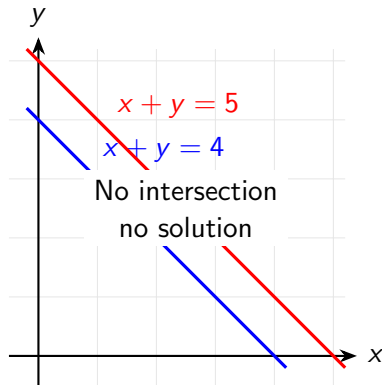
In two dimensions:

$$x + y = 4,$$

$$x + y = 5$$

has no solution.

The two equations describe parallel lines.



Consistent and Inconsistent Systems II

Consider the three equations

$$x = 1,$$

$$y = 1,$$

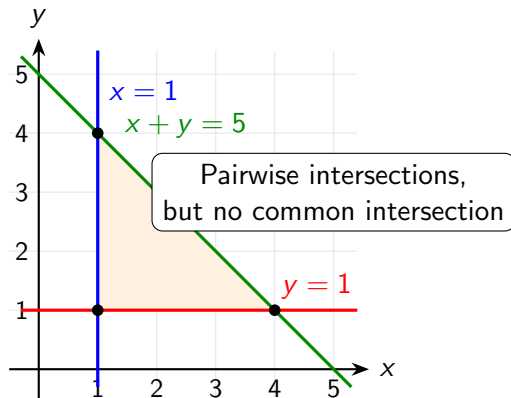
$$x + y = 5.$$

There is no single point that satisfies all three equations.

The first two equations require

$$(x, y) = (1, 1),$$

but this point does not satisfy $x + y = 5$.



Systems of Linear Equations

Consider

$$2x + y = 5,$$

$$x + 3y = 6.$$

The same information can be organised as

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 5 \\ 6 \end{bmatrix}}_{\mathbf{b}}.$$

So the compact form is

$$A\mathbf{x} = \mathbf{b}.$$

From Equations to Matrices

Consider three linear equations in three unknowns:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2,$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3.$$

The coefficients can be organised into a rectangular array:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{matrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Matrices

A matrix is a rectangular array of numbers with elements or entries arranged in rows and columns.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Matrices can represent:

- systems of equations,
- transformations of vectors,
- data tables,
- linear models.

Transpose

The transpose turns rows into columns and columns into rows.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

If

$$A \in \mathbb{R}^{m \times n},$$

then

$$A^T \in \mathbb{R}^{n \times m}.$$

A useful way to read vectors

A column vector becomes a row vector:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{x}^T = [x_1 \quad x_2 \quad x_3].$$

Why it matters

The dot product can be written as

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

Matrix-Vector Multiplication

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix}.$$

Each row of the matrix creates one component of the output vector.

Matrix–Vector Multiplication: One Row at a Time

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The result $A\mathbf{x}$ has one component for each row of A :

$$A\mathbf{x} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Step 1

Multiply the entries of the first row by the corresponding entries of $\mathbf{x}$, then add:

$$y_1 = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x_1 + x_2.$$

Matrix–Vector Multiplication: One Row at a Time

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The result $A\mathbf{x}$ has one component for each row of A :

$$A\mathbf{x} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Step 2

Repeat the same operation using the second row:

$$y_2 = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 + 3x_2.$$

Matrix–Vector Multiplication: One Row at a Time

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The result $A\mathbf{x}$ has one component for each row of A :

$$A\mathbf{x} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Step 3

Place the two results into the output vector:

$$A\mathbf{x} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix}.$$

Matrix–Vector Multiplication as a Linear Combination

The matrix can also be viewed as a collection of column vectors:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = [\mathbf{a}_1 \quad \mathbf{a}_2],$$

where

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2,$$

or, explicitly,

$$A\mathbf{x} = x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Matrix–Matrix Multiplication

Matrix–matrix multiplication means multiplying a matrix by several vectors at once. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Write B in terms of its columns:

$$B = [\mathbf{b}_1 \quad \mathbf{b}_2].$$

Then

$$AB = [A\mathbf{b}_1 \quad A\mathbf{b}_2].$$

Key idea

Each column of AB is obtained by applying A to the corresponding column of B .

Matrix–Matrix Multiplication: Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = [\mathbf{b}_1 \quad \mathbf{b}_2].$$

The columns are

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Therefore,

$$AB = [A\mathbf{b}_1 \quad A\mathbf{b}_2] = \begin{bmatrix} 2 & 5 \\ 1 & 5 \end{bmatrix}.$$

So

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & 5 \end{bmatrix}$$

Hadamard Product: Element-Wise Multiplication

The **Hadamard product** multiplies corresponding elements. The inputs must have the same dimensions.

Vectors

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{u} \odot \mathbf{v} = \begin{bmatrix} u_1 v_1 \\ u_2 v_2 \\ \vdots \\ u_n v_n \end{bmatrix}$$

$$\text{Example: } \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \odot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 18 \end{bmatrix}.$$

The result is a vector in $\mathbb{R}^n$.

Matrices

For $A, B \in \mathbb{R}^{m \times n}$,

$$(A \odot B)_{ij} = a_{ij} b_{ij}$$

Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \odot \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 12 \\ 21 & 32 \end{bmatrix}.$$

The result is an $m \times n$ matrix.

Matrices as Transformations

A matrix transforms an input vector $\mathbf{x}$ into an output vector:

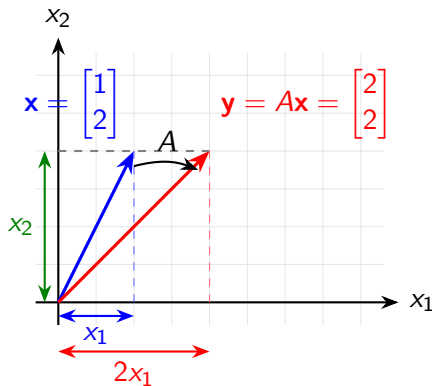
$$\mathbf{y} = A\mathbf{x}.$$

$$\text{when } A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

this gives

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix}.$$

The horizontal component is doubled, while the vertical component remains unchanged.



$$x_1 \mapsto 2x_1, \quad x_2 \mapsto x_2$$

Column Space: What Outputs Can a Matrix Produce?

Let

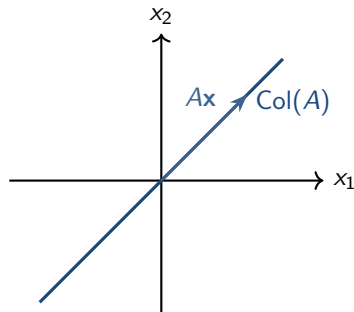
$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} = (x_1 + x_2) \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Every output lies on the same line.

$$\text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^2\} = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}.$$



Interpretation

The column space is the set of all possible outputs of the transformation. Its dimension is the rank of the matrix.

Null Space: Which Input Directions Disappear?

For the same matrix,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

the null space contains all inputs mapped to zero:

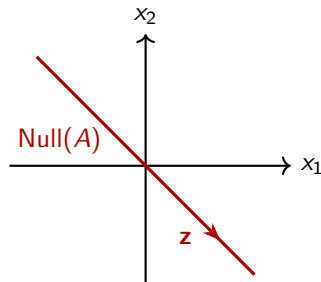
$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}.$$

Since

$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

we have

$$\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$



Interpretation

If $\mathbf{z} \in \text{Null}(A)$, then

$$A(\mathbf{x} + \mathbf{z}) = A\mathbf{x}.$$

Rank and the Rank–Nullity Theorem

For a matrix $A \in \mathbb{R}^{m \times n}$, the input space has dimension n .

Rank

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

The rank counts the number of independent output directions produced by A .

Nullity

$$\text{nullity}(A) = \dim(\text{Null}(A)).$$

The nullity counts the number of independent input directions that disappear.

Rank–nullity

$$\text{rank}(A) + \text{nullity}(A) = n$$

$$\text{For } A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

we have $\text{rank}(A) = 1$, $\text{nullity}(A) = 1$.

Geometric meaning

input dimensions = surviving directions + lost directions:
 $2 = 1 + 1$.

Determinant

Let A be an $n \times n$ matrix with real entries, written in terms of its columns as:

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n].$$

The determinant is a scalar-valued function

$$\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$$

uniquely characterised by three properties:

- 1 **Multilinearity:** it is linear in each column separately.
- 2 **Alternation:** swapping two columns changes the sign.
- 3 **Normalisation:** $\det(I) = 1$.

These properties imply

$$\det(A) = 0 \iff \mathbf{a}_1, \dots, \mathbf{a}_n \text{ are linearly dependent.}$$

Computing a 2×2 Determinant

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant may be written as

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \implies \det(A) = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 2(3) - 1(1) = 5.$$

Since $\det(A) \neq 0$, the columns of A are linearly independent.

Computing Determinants in Higher Dimensions

For larger square matrices, determinants can be computed by expanding along a row or column.

For a 3×3 matrix,

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

expand along the first row:

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Using the 2×2 formula inside each minor,

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Determinant and Invertibility

$$\det(A) \neq 0 \iff \begin{array}{l} \text{columns are linearly independent} \\ \text{rank}(A) = n \\ A^{-1} \text{ exists} \end{array}$$

When $\det(A) = 0$:

- the columns are linearly dependent;
- at least one direction is lost;
- the transformation cannot be reversed uniquely.

In $\mathbb{R}^2$, the plane collapses onto a line or a point.

The Inverse Matrix

The inverse matrix reverses the action of A .

$$A\mathbf{x} = \mathbf{b}$$

If A^{-1} exists, then

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}.$$

Since

$$A^{-1}A = I,$$

we get

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

So the inverse gives a direct way to recover $\mathbf{x}$ from $\mathbf{b}$.

Computing the Inverse of a 2×2 Matrix

$$\text{For } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if

$$\det(A) = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Rule

Swap the diagonal entries, change the signs of the off-diagonal entries, and divide by the determinant.

Inverse Matrix: Worked Example

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

First compute the determinant:

$$\det(A) = 2(3) - 1(1) = 5.$$

Since $\det(A) \neq 0$, the inverse exists.

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}.$$

Using the Inverse to Solve a System

For the system

$$A\mathbf{x} = \mathbf{b},$$

if A^{-1} exists, then

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

Using

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix},$$

we get

$$\mathbf{x} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 9 \\ 7 \end{bmatrix} = \begin{bmatrix} 9/5 \\ 7/5 \end{bmatrix}.$$

Computing Inverses by Row Reduction

For larger matrices, the inverse is usually computed using row reduction:

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

There are only three elementary row operations:

$$R_i \leftrightarrow R_j, \quad R_i \leftarrow cR_i \quad (c \neq 0), \quad R_i \leftarrow R_i + cR_j.$$

These correspond to swapping two rows, scaling a row, and adding a multiple of one row to another.

The procedure is:

- place A beside the identity matrix I ;
- apply the same row operations to both sides;
- continue until the left side becomes I ;

The inverse exists only if A can be row-reduced to I .

Row Reduction Example

$$\text{Start with } A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Form the augmented matrix:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right].$$

After row reduction,

$$\left[\begin{array}{cc|cc} 1 & 0 & 3/5 & -1/5 \\ 0 & 1 & -1/5 & 2/5 \end{array} \right].$$

Therefore,

$$A^{-1} = \begin{bmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{bmatrix}.$$

Dot Product

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n.$$

Using the transpose, the same operation can be written as

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}}$$

because

$$\mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + \cdots + u_n v_n.$$

The dot product produces a scalar.

Dot Product: Example

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 1(2) + 2(-1) = 0.$$

Vector Norms: Measuring the Size of a Vector

A **norm** assigns a non-negative size to a vector: $\mathbf{x} \mapsto \|\mathbf{x}\|$.

A norm must satisfy four properties.

1. Non-negativity

$$\|\mathbf{x}\| \geq 0.$$

A vector cannot have a negative size.

2. Zero vector

$$\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}.$$

Only the zero vector has zero size.

3. Scaling

$$\|a\mathbf{x}\| = |a| \|\mathbf{x}\|.$$

Scaling a vector by a scales its size by $|a|$.

4. Triangle inequality

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

The direct route is shorter or equal

The Euclidean Norm

The most familiar norm is the **Euclidean norm**, or 2-norm:

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Recall the dot product

$$\mathbf{x} \cdot \mathbf{x} = x_1^2 + x_2^2 + \cdots + x_n^2.$$

Therefore,

$$\boxed{\|\mathbf{x}\|_2 = \sqrt{\mathbf{x} \cdot \mathbf{x}}}$$

The Euclidean norm gives the ordinary geometric length of a vector.

Example

Let

$$\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

First compute the dot product:

$$\mathbf{x}^T \mathbf{x} = \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3^2 + 4^2 = 25.$$

Hence,

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{25} = 5.$$

Different Norms Measure Size Differently

The same vector can be assigned different sizes depending on the norm used.

$$\mathbf{x} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}.$$

1-norm

$$\begin{aligned}\|\mathbf{x}\|_1 &= \sum_{i=1}^n |x_i| \\ &= |-2| + |1| + |3| \\ &= 6.\end{aligned}$$

Adds the magnitude of every component.

∞ -norm

$$\begin{aligned}\|\mathbf{x}\|_\infty &= \max_i |x_i| \\ &= \max\{2, 1, 3\} \\ &= 3.\end{aligned}$$

Measures only the largest component magnitude.

Orthogonality and Orthonormal Bases

Two vectors are orthogonal when their dot product is zero:

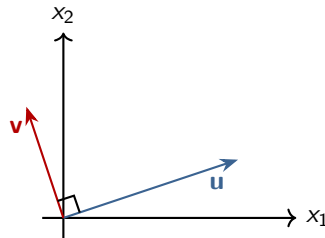
$$\mathbf{u} \perp \mathbf{v} \quad \Longleftrightarrow \quad \mathbf{u}^T \mathbf{v} = 0.$$

Orthonormal vectors

Vectors $\mathbf{q}_1, \dots, \mathbf{q}_k$ are orthonormal if

$$\mathbf{q}_i^T \mathbf{q}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

They are mutually perpendicular and each has unit length.



For an orthonormal basis $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_n]$,

we have

$$Q^T Q = I, \quad Q^{-1} = Q^T.$$

Projection: Geometric Idea

Projection asks:

What part of $\mathbf{x}$ lies in the direction of $\mathbf{u}$?

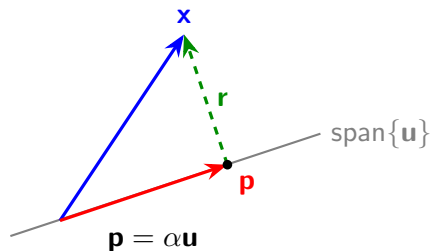
The projection is a vector $\mathbf{p}$ lying along $\mathbf{u}$.

$$\mathbf{p} = \alpha \mathbf{u}.$$

The leftover part

$$\mathbf{r} = \mathbf{x} - \mathbf{p}$$

is perpendicular to $\mathbf{u}$.



Projection Formula

Since $\mathbf{r} = \mathbf{x} - \mathbf{p}$

is orthogonal to $\mathbf{u}$,

$$(\mathbf{x} - \alpha\mathbf{u}) \cdot \mathbf{u} = 0.$$

Expanding gives $\mathbf{x} \cdot \mathbf{u} - \alpha(\mathbf{u} \cdot \mathbf{u}) = 0.$

Therefore,

$$\alpha = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}.$$

So

$$\text{proj}_{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

Eigenvectors: Directions That Keep Their Direction

A matrix usually changes both the length and the direction of a vector.

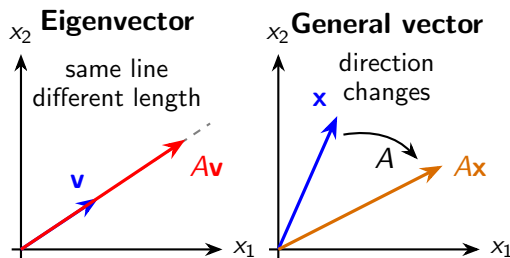
However, some special vectors remain on the same line after the transformation:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

These vectors are called **eigenvectors**.

The scalar λ is the corresponding **eigenvalue**:

- $|\lambda| > 1$: the vector is stretched;
- $0 < |\lambda| < 1$: the vector is shortened;
- $\lambda < 0$: its direction is reversed.



An eigenvector may change length, but remains on the same direction.

Definition of Eigenvalues and Eigenvectors

Let A be a square matrix.

A nonzero vector $\mathbf{v}$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar λ .

Equivalently,

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Important point

The vector $\mathbf{v}$ must be nonzero. Otherwise every value of λ would satisfy the equation trivially.

Simple Example: A Stretching Matrix

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Then

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

So the coordinate axes are eigenvector directions:

$\mathbf{e}_1$ has eigenvalue 2, $\mathbf{e}_2$ has eigenvalue 3.

Finding Eigenvalues

Starting from

$$(A - \lambda I)\mathbf{v} = \mathbf{0},$$

we need a nonzero solution $\mathbf{v}$.

This is possible only when the matrix $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0.$$

This equation is called the **characteristic equation**.

Procedure

- 1 Solve $\det(A - \lambda I) = 0$ to find the eigenvalues.
- 2 For each eigenvalue, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ to find the eigenvectors.

Eigenvalue Example

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}.$$

The characteristic equation is

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0.$$

Therefore,

$$(2 - \lambda)^2 = 1 \quad \implies \quad \lambda = 3 \quad \text{or} \quad \lambda = 1.$$

Eigenvectors for the Example

$$\text{For } A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

we found the eigenvalues $\lambda = 3$ and $\lambda = 1$.

For $\lambda = 3$:

$$(A - 3I)\mathbf{v} = 0 \quad \implies \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda = 1$:

$$(A - I)\mathbf{v} = 0 \quad \implies \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The matrix stretches the direction $[1, 1]^T$ by 3, and the direction $[1, -1]^T$ by 1.

Diagonalisation: Using Eigenvectors as Coordinates

If a matrix has enough independent eigenvectors, we can collect them in a matrix

$$P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n] .$$

Then

$$A = PDP^{-1},$$

$$\text{where, } D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} .$$

In eigenvector coordinates, A acts as independent scalings.

Symmetric Matrices and Orthogonal Eigenvectors

A square matrix is symmetric when

$$A = A^T.$$

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Symmetric matrices have two important properties:

- all eigenvalues are real;
- their eigenvectors can be chosen orthonormally.

Collect the orthonormal eigenvectors in

$$Q = [\mathbf{q}_1 \quad \cdots \quad \mathbf{q}_n].$$

Then

$$A = Q\Lambda Q^T, \quad \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

Because $Q^{-1} = Q^T$, this is a particularly simple form of diagonalisation.

Data as a Matrix

Many datasets can be organised as a matrix:

$$X = \begin{bmatrix} - & \mathbf{x}_1^T & - \\ - & \mathbf{x}_2^T & - \\ & \vdots & \\ - & \mathbf{x}_m^T & - \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}.$$

Rows

Each row is one observation, sample, experiment, image, patient, time point, or operating condition.

Columns

Each column is one feature, measured variable, sensor, pixel, concentration, temperature, or descriptor.

A Small Data Matrix Example

Suppose we measure three variables from four experiments:

$$X = \begin{bmatrix} 20 & 5.1 & 101 \\ 25 & 5.8 & 108 \\ 30 & 6.4 & 115 \\ 35 & 7.0 & 121 \end{bmatrix}.$$

Here:

- there are $m = 4$ observations;
- there are $n = 3$ features;
- the dataset is one point cloud in $\mathbb{R}^3$.

	f1	f2	f3
obs. 1			
obs. 2			
obs. 3			
obs. 4			

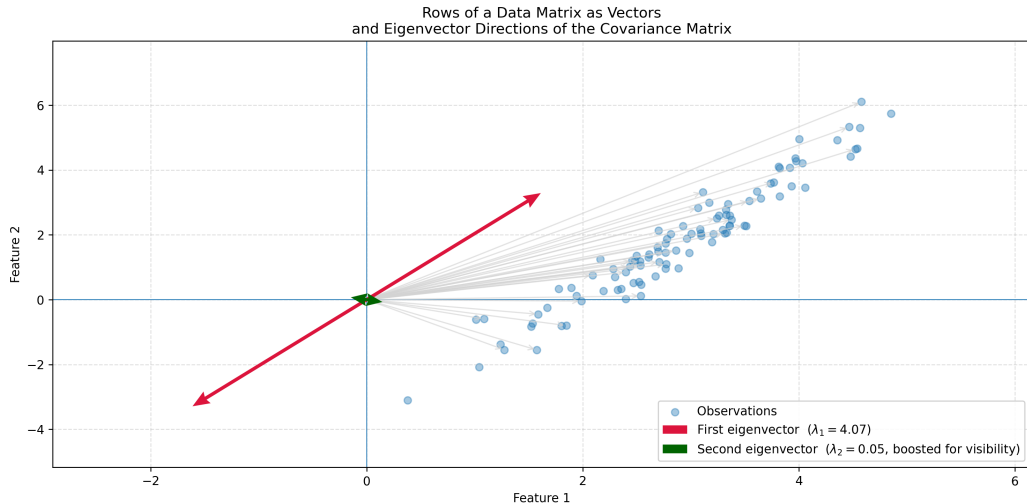
one row = one vector

one column = one measured variable

$$\mathbf{x}_i \in \mathbb{R}^3$$

Each observation is a vector of features.

Eigenvectors and Eigenvalues of Data



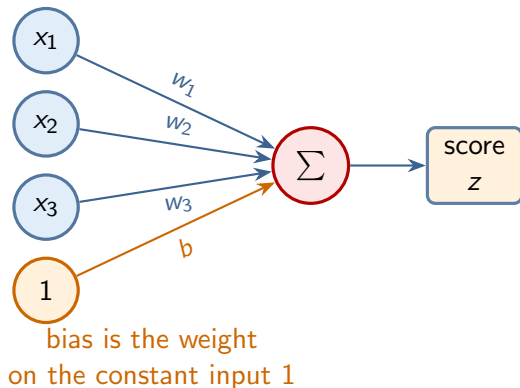
Basic Building Block for a Linear Model: A Weighted Sum

Let the input and weight vectors be

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix}.$$

Their dot product gives the weighted sum:

$$\mathbf{w}^T \mathbf{x} = \begin{bmatrix} w_1 & w_2 & w_3 & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}.$$



$$z = \mathbf{w}^T \mathbf{x} = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

Several Weighted Sums Become a Matrix–Vector Product

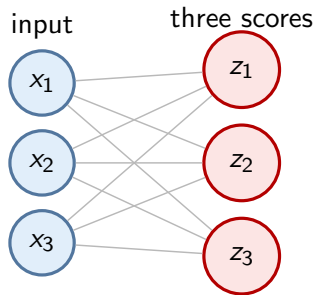
Suppose we want several scores from the same input.

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}.$$

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} - & \mathbf{w}_1^T & - \\ - & \mathbf{w}_2^T & - \\ - & \mathbf{w}_3^T & - \end{bmatrix} \mathbf{x} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Key point

Each row of $\mathbf{W}$ is one weighted sum.



bias vector
 $\mathbf{b} = [b_1, b_2, b_3]^T$

Why Matrix–Matrix Multiplication Matters in Machine Learning

Machine-learning models usually process a batch of input vectors together:

$$\mathbf{X} = \begin{bmatrix} - & \mathbf{x}_1^T & - \\ - & \mathbf{x}_2^T & - \\ & \vdots & \\ - & \mathbf{x}_B^T & - \end{bmatrix}.$$

The same layer is then evaluated using one matrix multiplication:

$$\mathbf{Z} = \mathbf{X}\mathbf{W}^T + \mathbf{1}\mathbf{b}^T$$

$$\underbrace{\mathbf{X}}_{B \times n} \underbrace{\mathbf{W}^T}_{n \times m} = \underbrace{\mathbf{Z}}_{B \times m}.$$

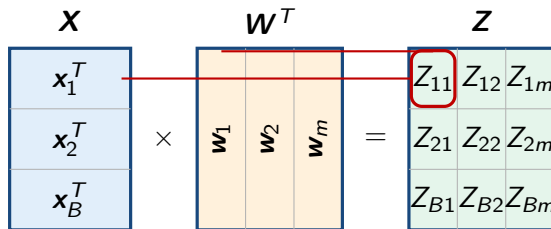
One matrix multiplication evaluates every neuron in the layer for every sample in the batch.

What Is Inside $\mathbf{XW}^T$?

The matrix product contains all sample–neuron dot products:

$$\mathbf{Z} = \mathbf{XW}^T.$$

$$Z_{ij} = \mathbf{x}_i^T \mathbf{w}_j.$$



Interpretation

Entry Z_{ij} is the score of weighted sum j applied to data sample i .

A Neural-Network Layer Is Still a Matrix Operation

A layer usually has two steps:

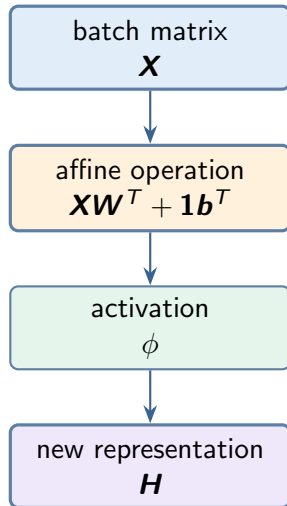
$$\mathbf{Z} = \mathbf{XW}^T + \mathbf{1b}^T,$$

$$\mathbf{H} = \phi(\mathbf{Z}).$$

Here ϕ is applied element by element.

Teaser

The activation function is what makes neural networks more than just one large linear calculation.



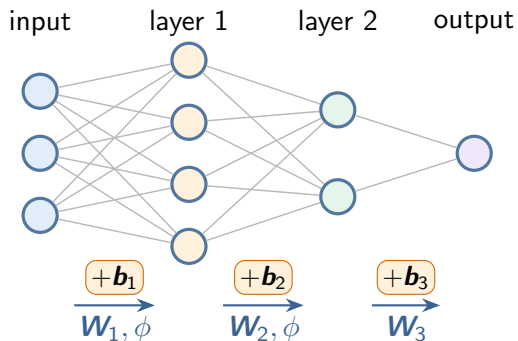
A Deep Network Repeats the Same Pattern

Each layer takes a representation and produces a new one:

$$\mathbf{H}_1 = \phi(\mathbf{X}\mathbf{W}_1^T + \mathbf{1}\mathbf{b}_1^T),$$

$$\mathbf{H}_2 = \phi(\mathbf{H}_1\mathbf{W}_2^T + \mathbf{1}\mathbf{b}_2^T),$$

$$\hat{\mathbf{Y}} = \mathbf{H}_2\mathbf{W}_3^T + \mathbf{1}\mathbf{b}_3^T.$$



The structure looks complicated, but the repeated building block is familiar linear algebra.

Why Large Matrix Operations Are Useful

For

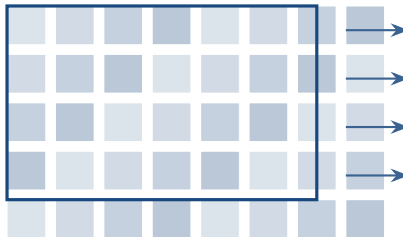
$$\mathbf{C} = \mathbf{AB}, \quad C_{ij} = \sum_{\ell} A_{i\ell} B_{\ell j},$$

different entries, rows, columns, or blocks of $\mathbf{C}$ can be computed at the same time.

Computational message

Machine learning becomes practical because many small dot products can be organised as a few large matrix products.

many entries of $\mathbf{C}$



computed in
parallel blocks

Final Picture: Linear Algebra in Data and Machine Learning

Data samples are represented as vectors and collected in matrices. Linear algebra then contributes in two closely related ways.

Computing with data

- Weighted sums are dot products.
- Matrix–vector products compute several outputs at once.
- Matrix–matrix products process several samples at once.
- Neural networks repeat these operations layer after layer.

Understanding the geometry of data

- Data occupy a high-dimensional vector space.
- Their structure may resemble a lower-dimensional subspace or manifold.
- Projections provide lower-dimensional representations.
- Eigenvectors identify important directions, as in PCA.

Linear algebra provides both the computations used by machine-learning models and the geometric language used to understand structure in data.